

VEDAANG CHOPRA

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MS CS (ML) at Georgia Tech — Advised by **Dr. Matthew Gombolay**, CORE Robotics Lab

Research: Verifiable-reward alignment on frozen models – Evaluation-driven reasoning systems – Production ML (4.5 yrs)

RESEARCH EXPERIENCE

Georgia Institute of Technology – CORE Robotics Lab @ Siemens

Atlanta, GA

Graduate Research Assistant — Advisor: Dr. Matthew Gombolay

Jan 2026 – Present

- **CAD Code Generation Pipeline:** Designing synthesis as inference-time search with verifiable geometric reward on a frozen model – reward signals from compile validity, geometric correctness (Chamfer/Hausdorff on STL/B-Rep), and structural diffs guide a LangGraph repair loop, an explicit alternative to training-time alignment for code generation; 16-module system maintained as a 202-test automated evaluation harness with VLM-based visual judgment.
- Designing verifier components that turn engineering constraints into checkable reward signals: failure classification, AST/code-graph structural analysis, and repair-trajectory quality scoring – correctness is checked against ground truth rather than asserted.

Georgia Institute of Technology – CS 8903 Special Problems

Atlanta, GA

Graduate Researcher

Jan 2025 – Present

- **ATHENA:** Built a 5-agent supervisor-worker system with reflection-based critique loops improving output quality measurably (+0.18 BLEU-4 absolute over single-agent baseline); designed cross-modal evaluation framework (BLEU-4, METEOR, BERTScore, CLIPScore, SSIM, PSNR) over 100 reference videos; ablation showed removing the critique agent cut fact coverage 27% – hallucination-style failures surfaced by systematic evaluation.

INDUSTRY EXPERIENCE

Fortinet Technologies Inc. (AIOps R&D)

Bengaluru, India

Software Development Engineer I & II (ML/AI)

Feb 2021 – Jul 2025

- Led development of an agentic RAG diagnostics system from hackathon prototype to production deployment: LLM plans multi-step tool calls over network telemetry and verifies hypotheses via structured function calling before concluding – verification-gated autonomous root-cause analysis reducing mean resolution time 70%.
- Trained cross-functional alignment stack with PyTorch DDP + mixed precision on H100/A100 clusters for encoder-frozen multimodal retrieval (Georgia Tech research); deployed edge ML inference via ONNX Runtime (40% latency reduction); scaled OpenSearch telemetry ingestion 50→2,000 events/sec (40x) in Golang.
- Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection; reduced manual troubleshooting 75%. Built SLA forecasting for 60+ classifiers across 4 categories with automated evaluation and retraining.

TECHNICAL SKILLS

Model Alignment & Evaluation: Verifiable-Reward Design & Verifiers, Inference-Time Search on Frozen Models, Model Evaluation & Benchmarking, Ablation Studies, LLM-as-a-Judge, Hallucination Detection (coursework), Calibration Concepts (coursework), Familiarity with Current Alignment & Training-Time Method Literatures (coursework/self-study)

LLMs / Reasoning: PyTorch, Hugging Face Transformers, Deep Learning, Transformers & Attention Architectures, Data Quality & Deduplication Pipelines, Structured Generation (Pydantic)

Inference & Systems: vLLM, Inference Optimization (ONNX Runtime), Latency/Throughput Engineering, Distributed Training (PyTorch DDP, Mixed Precision), Quantization Contexts (ONNX), Python, Go, SQL/PostgreSQL, OpenSearch/Elasticsearch, Docker, Linux, Git

Agentic AI: LangChain, LangGraph, Agentic RAG, Tool/Function Calling, Multi-Agent Orchestration, Long-Horizon Task Decomposition, Reflection & Critique Loops

Reinforcement Learning: PPO, DQN, Reward Shaping, Curriculum Learning, Self-Play (academic projects)

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. in Computer Science (Machine Learning Specialization), **GPA: 4.0/4.0**

Aug 2024 – Dec 2026

- **Coursework:** Deep Learning, Deep RL, ML Security, Large & Vision Language Models, Agentic AI, Systems for AI

Maharaja Surajmal Institute of Technology

New Delhi, India

B.Tech. in Information Technology, **CGPA: 8.8/10.0**

Aug 2016 – Aug 2020